

PROBLEM SET 4—SOLUTIONS

ECON301—ECONOMETRICS I

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These are my suggested solutions. It is your responsibility to cross-check this document with your own answers.

Note: in the lecture notes and videos, I usually denote vectors by underlining them, for example $\underline{x}$, and I denote matrices in capital letters. When typesetting it, I use **bold notation and upright letters** to denote vectors and matrices, i.e., $\mathbf{x}$ and $\mathbf{M}$ for example.

1. Consider the normal equations $(\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y}$; the previous matrix equation was derived from the system of first order conditions, if we “unpack it” we have the following linear system of equations:

$$\begin{aligned} \hat{\beta}_0 n + \hat{\beta}_1 \sum_i X_{i1} + \cdots + \hat{\beta}_k \sum_i X_{ik} &= \sum_i Y_i \\ \hat{\beta}_0 \sum_i X_{i1} + \hat{\beta}_1 \sum_i X_{i1}^2 + \cdots + \hat{\beta}_k \sum_i X_{ik} X_{i1} &= \sum_i Y_i X_{i1} \\ &\vdots \\ \hat{\beta}_0 \sum_i X_{ik} + \hat{\beta}_1 \sum_i X_{i1} X_{ik} + \cdots + \sum_i X_{ik}^2 &= \sum_i Y_i X_{ik} \end{aligned} \quad (1)$$

The system (1) has $k + 1$ equations in $k + 1$ unknowns.

Take the first equation in (1) and solve it for $\hat{\beta}_0$ and simplify to get:

$$\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{X}_1 - \cdots - \hat{\beta}_k \bar{X}_k \quad (2)$$

Substitute equation (2) into the system (1) and simplify to get:

$$\begin{aligned} \hat{\beta}_1 \left(\sum_i X_{i1}^2 - n \bar{X}_1^2 \right) + \cdots + \hat{\beta}_k \left(\sum_i X_{ik} X_{i1} - n \bar{X}_1 \bar{X}_k \right) &= \sum_i Y_i X_{i1} - n \bar{Y} \bar{X}_1 \\ &\vdots \\ \hat{\beta}_1 \left(\sum_i X_{i1} X_{ik} - n \bar{X}_1 \bar{X}_k \right) + \cdots + \hat{\beta}_k \left(\sum_i X_{ik}^2 - n \bar{X}_k^2 \right) &= \sum_i Y_i X_{ik} - n \bar{Y} \bar{X}_k \end{aligned} \quad (3)$$

The system (3) is a system of k equations in k variables. By the algebraic results in lemma 3.1, the system (3) can be expressed as

$$\begin{aligned} \hat{\beta}_1 S_{x_1, x_1} + \cdots + \hat{\beta}_k S_{x_1, x_k} &= S_{x_1, y} \\ &\vdots \\ \hat{\beta}_1 S_{x_k, x_1} + \cdots + \hat{\beta}_k S_{x_k, x_k} &= S_{x_k, y} \end{aligned} \quad (4)$$

Note that the matrix $\mathbf{S}_{xx} = [S_{x_i, x_j}]_{i=1, \dots, k, j=1, \dots, k}$ and the column vector $\mathbf{S}_{xy} = [S_{x_i, y}]_{i=1, \dots, k}$. Therefore, the system (4) can be expressed in matrix form as

$$\mathbf{S}_{xx} \hat{\boldsymbol{\beta}}_P = \mathbf{S}_{xy}. \quad (5)$$

2. There are several ways to prove this. In here I present two. The first one is a direct proof, and the second uses the idea of projection matrices.

Method 1—direct proof: Consider the fitted values

$$\hat{Y}_i = \hat{\beta}_0 + \hat{\beta}_1 X_{i1} + \cdots + \hat{\beta}_k X_{ik} \quad i = 1, \dots, n.$$

Add all the equations above and divide by n to get

$$\bar{\hat{Y}} = \frac{1}{n} \sum_i \hat{Y}_i = \hat{\beta}_0 + \hat{\beta}_1 \bar{X}_1 + \cdots + \hat{\beta}_k \bar{X}_k.$$

Now, use equation (2) from task 1 to get

$$\widehat{Y} = \bar{Y} - \hat{\beta}_1 \bar{X}_1 - \dots - \bar{X}_k + \hat{\beta}_1 \bar{X}_1 + \dots + \hat{\beta}_k \bar{X}_k = \bar{Y}.$$

Similarly, consider the errors

$$\hat{\epsilon}_i = Y_i - \widehat{Y}_i.$$

Add all equations above and divide by n to get

$$\bar{\epsilon} = \frac{1}{n} \sum_i \hat{\epsilon}_i = \bar{Y} - \widehat{Y} = 0,$$

where the last equation comes from the result that $\widehat{Y} = \bar{Y}$.

Method 2—projection matrices: Consider the projection matrices $\mathbf{P}_1$ and $\mathbf{P}_X$ as defined in lecture 5. Since $\mathbf{1} \in \mathcal{L}(\mathbf{X})$ then $\mathbf{P}_X \mathbf{1} = \mathbf{1}$. Note that $\mathbf{P}_1 = \mathbf{1}(\mathbf{1}^\top \mathbf{1})^{-1} \mathbf{1}^\top = (1/n) \mathbf{1} \mathbf{1}^\top$. Thus $\mathbf{P}_1 \mathbf{P}_X = (1/n) \mathbf{1} \mathbf{1}^\top \mathbf{P}_X = (1/n) \mathbf{1} (\mathbf{P}_X \mathbf{1})^\top = (1/n) \mathbf{1} \mathbf{1}^\top = \mathbf{P}_1$, where the second equality uses a property of transposition¹ and the symmetry of $\mathbf{P}_X$. Then $\widehat{\mathbf{Y}} \mathbf{P}_1 \widehat{\mathbf{Y}} = \mathbf{P}_1 \mathbf{P}_X \mathbf{Y} = \mathbf{P}_1 \mathbf{Y} = \bar{Y} \mathbf{1}$, i.e., a vector of dimension n with $\bar{Y}$ in every component. On the other hand $\mathbf{P}_1 \mathbf{N}_X = \mathbf{P}_1 (\mathbf{I} - \mathbf{P}_X) = \mathbf{P}_1 - \mathbf{P}_1 \mathbf{P}_X = \mathbf{P}_1 - \mathbf{P}_1 = \mathbf{O}$, where $\mathbf{O}$ is a $n \times n$ matrix full of zeros. Hence $\mathbf{P}_1 \hat{\epsilon} = \mathbf{P}_1 \mathbf{N}_X \mathbf{Y} = \mathbf{O} \mathbf{Y} = \mathbf{0}$.

3. Because of assumptions 1M (random sampling), we have that the following results:

$$\frac{1}{n} \sum_{i=1}^n Y_i^2 \rightarrow_P E[Y_i^2], \quad (6)$$

$$\frac{1}{n} \sum_{i=1}^n Y_i \rightarrow_P E[Y_i], \quad (7)$$

$$\frac{1}{n} \sum_{i=1}^n Y_i X_i \rightarrow_P E[Y_i X_i], \quad (8)$$

since each one of the expressions above is a sample moment, and by the Weak Law of Large Numbers, all of them converge in probability to the corresponding population moment. The case of $\frac{1}{n} \sum_{i=1}^n Y_i^2$ requires a little bit of care, but we have done this before. Check the example in page 16 of Lecture 4 to see that since Y_i and Y_j are independent, then Y_i^2 and Y_j^2 are also independent, and therefore, their covariance is zero. This implies that the hypothesis from theorem 4.4 (WLLN) hold and we can conclude the result above. Similarly, for the case of $\frac{1}{n} \sum_{i=1}^n Y_i X_i$, the fact that (Y_i, X_i) is independent from (Y_j, X_j) implies that $X_i Y_i$ and $X_j Y_j$ are also independent, and therefore, their covariance is zero. Again, this implies that the hypothesis from theorem 4.4 (WLLN) hold and we can conclude the result above.

Furthermore, assumptions 1M and 2M yield the consistency of the LSE (Thm 5.4): $\hat{\beta} = (\hat{\beta}_0, \hat{\beta}_1)^\top \rightarrow_P \beta = (\beta_0, \beta_1)^\top$. Therefore, an application of Theorem 4.3 from lecture 4 yields:

$$\begin{aligned} \frac{1}{n} \sum_{i=1}^n \hat{\beta}_0 Y_i &= \hat{\beta}_0 \left(\frac{1}{n} \sum_{i=1}^n Y_i \right) \rightarrow_P \beta_0 E[Y_i], \\ \frac{1}{n} \sum_{i=1}^n \hat{\beta}_1 Y_i X_i &= \hat{\beta}_1 \left(\frac{1}{n} \sum_{i=1}^n Y_i X_i \right) \rightarrow_P \beta_1 E[Y_i X_i], \end{aligned}$$

4. It remains to show statements (ii), (iii), and (iv) from Theorem 5.2.

(ii) We need to show that $\mathbf{N}_X = \mathbf{I} - \mathbf{P}_X$ is a projection matrix, so we need to show that it is symmetric and idempotent. *Symmetry:* $(\mathbf{I} - \mathbf{P}_X)^\top = \mathbf{I}^\top - \mathbf{P}_X^\top = \mathbf{I} - \mathbf{P}_X$ since the identity matrix is symmetric and $\mathbf{P}_X$ is symmetric from statement (i) in the theorem. *Idempotency:* $(\mathbf{I} - \mathbf{P}_X)(\mathbf{I} - \mathbf{P}_X) = \mathbf{I} - \mathbf{P}_X - \mathbf{P}_X + \mathbf{P}_X \mathbf{P}_X = \mathbf{I} - \mathbf{P}_X - \mathbf{P}_X + \mathbf{P}_X$, where for the last term we have used the fact that $\mathbf{P}_X$ is idempotent from statement (i) in the theorem. Therefore, the last term is equal to $\mathbf{I} - \mathbf{P}_X = \mathbf{N}_X$.

¹(AB)[⊤] = B[⊤]A[⊤].

(iii) $P_X X = (X(X^T X)^{-1} X^T) X = X$.

(iv) $N_X X = (I - P_X) X = IX - P_X X = X - X = O$, where in the last equality we used the result from (iii) above.

5. Consider the problem

$$SS(c_0, c_1, \dots, c_k, c_{k+1}) = \frac{1}{n} \sum_{i=1}^n (y_i - c_0 - c_1 x_{i,1} - c_k x_{i,k} - c_{k+1} x_{i,k+1})^2. \quad (9)$$

It is clear that

$$\frac{SSE(X_1, \dots, X_k, X_{k+1})}{n} = SS(\hat{\beta}_0, \hat{\beta}_1, \dots, \hat{\beta}_k, \hat{\beta}_{k+1}) = \min_{(c_0, c_1, \dots, c_k, c_{k+1})} SS(c_0, c_1, \dots, c_k, c_{k+1}), \quad (10)$$

where $(\hat{\beta}_0, \hat{\beta}_1, \dots, \hat{\beta}_k, \hat{\beta}_{k+1})$ is the LSE coefficient vector from the regression with $k + 1$ explanatory variables. Consider now the constrained problem

$$\begin{aligned} \min_{(c_0, c_1, \dots, c_k, c_{k+1})} & SS(c_0, c_1, \dots, c_k, c_{k+1}) \\ \text{subject to: } & c_{k+1} = 0. \end{aligned} \quad (11)$$

It is clear that the constrained problem (11) has a minimized value equal to

$$SS(\hat{\beta}_0, \hat{\beta}_1, \dots, \hat{\beta}_k) = \frac{SSE(X_1, \dots, X_k)}{n}, \quad (12)$$

where $(\hat{\beta}_0, \hat{\beta}_1, \dots, \hat{\beta}_k)$ is the LSE coefficient vector from the regression with k explanatory variables.²

Therefore, it is clear that the minimized value of problem (10) is smaller than the minimized value of problem (11) because of the extra constraint in the later one.

Comparing equations (10) and (12) we obtain that

$$\frac{SSE(X_1, \dots, X_k, X_{k+1})}{n} = SS(\hat{\beta}_0, \hat{\beta}_1, \dots, \hat{\beta}_k, \hat{\beta}_{k+1}) \leq SS(\hat{\beta}_0, \hat{\beta}_1, \dots, \hat{\beta}_k) = \frac{SSE(X_1, \dots, X_k)}{n},$$

which implies that $SSE(X_1, \dots, X_k, X_{k+1}) \leq SSE(X_1, \dots, X_k)$. The result from this exercise implies that the R^2 from the regression with $k + 1$ variables is larger than the one with k variables.

²Note that $(\hat{\beta}_0, \hat{\beta}_1, \dots, \hat{\beta}_k, \hat{\beta}_{k+1})$ and $(\hat{\beta}_0, \hat{\beta}_1, \dots, \hat{\beta}_k)$ do not necessarily have the same first k components.